



What are LLM-Powered Autonomous Agents? - TruEra

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Imagine a world where your artificial intelligence (AI) not only understands your requests but also acts as a knowledgeable virtual assistant, organizes your agenda, helps you make complex business decisions, or even accelerates scientific research. That is the ambition of researchers and industry actors working on developing large language model (LLM)-powered autonomous agents such as [AutoGPT](#), [BabyAGI](#) and [GPT-Engineer](#). These autonomous agents are intelligent systems that rely on large language models as their core controllers to achieve specific goals given in natural language. In this article, we will dive into the realm of LLM-powered autonomous agents, explaining what they are, potential applications.

Understanding an Agent System

In a LLM-powered autonomous agent system, the LLM functions as the agent's brain, complemented by three components: planning, memory and tool use. Let's review these components in detail.

Part 1: Planning

One of the defining features of an AI agent is its ability to execute a plan. To this end, it needs to be able to decompose a task and improve through self-reflection. Task decomposition refers to an agent's ability to break down complex tasks into smaller, manageable subgoals. To do so, AI developers often use [Chain of Thought](#) (a series of intermediate natural language reasoning steps that lead to the final output) and [Tree of Thoughts](#) (each thought is a coherent language sequence that serves as an intermediate step toward problem solving) techniques. Once combined, these methods prompt the model to "think step by step" and decompose hard tasks into simpler steps. Some agents rely on external classical planners to perform long-horizon planning, using the Planning Domain Definition Language (PDDL) method to translate the problem into a format that external planners can understand and generate plans for.

In contrast, self-reflection allows AI agents to improve iteratively [by refining past action decisions and correcting previous mistakes](#). To improve agents' self-reflection capabilities, AI developers increasingly use [ReAct](#) (a versatile approach that integrates language models with reasoning and action)) and [reflexion](#) (an innovative framework that enhances the learning and decision-making capabilities of language agents by leveraging linguistic

